CS 340: Algorithms and Data Structures

AVL Trees

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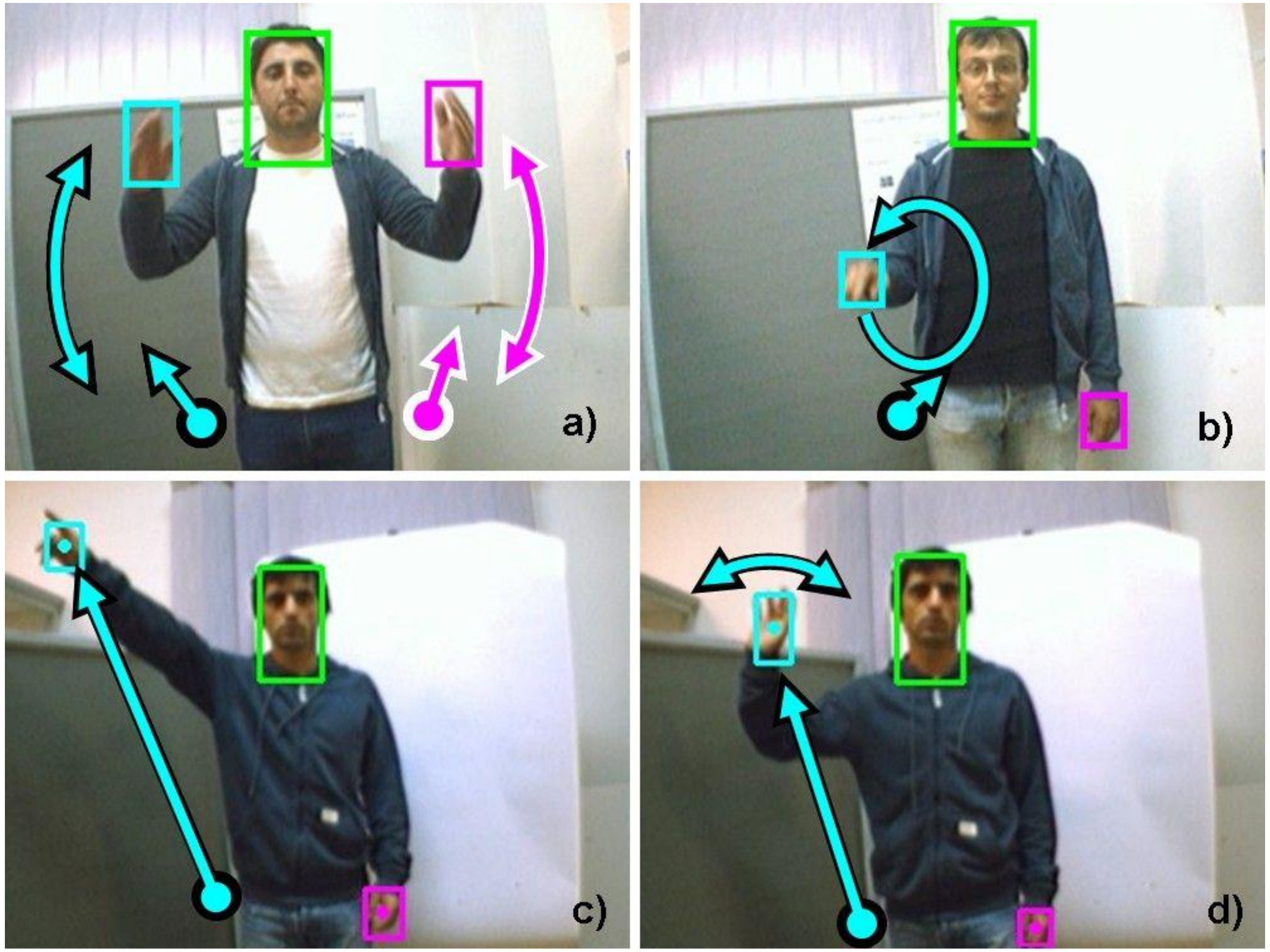
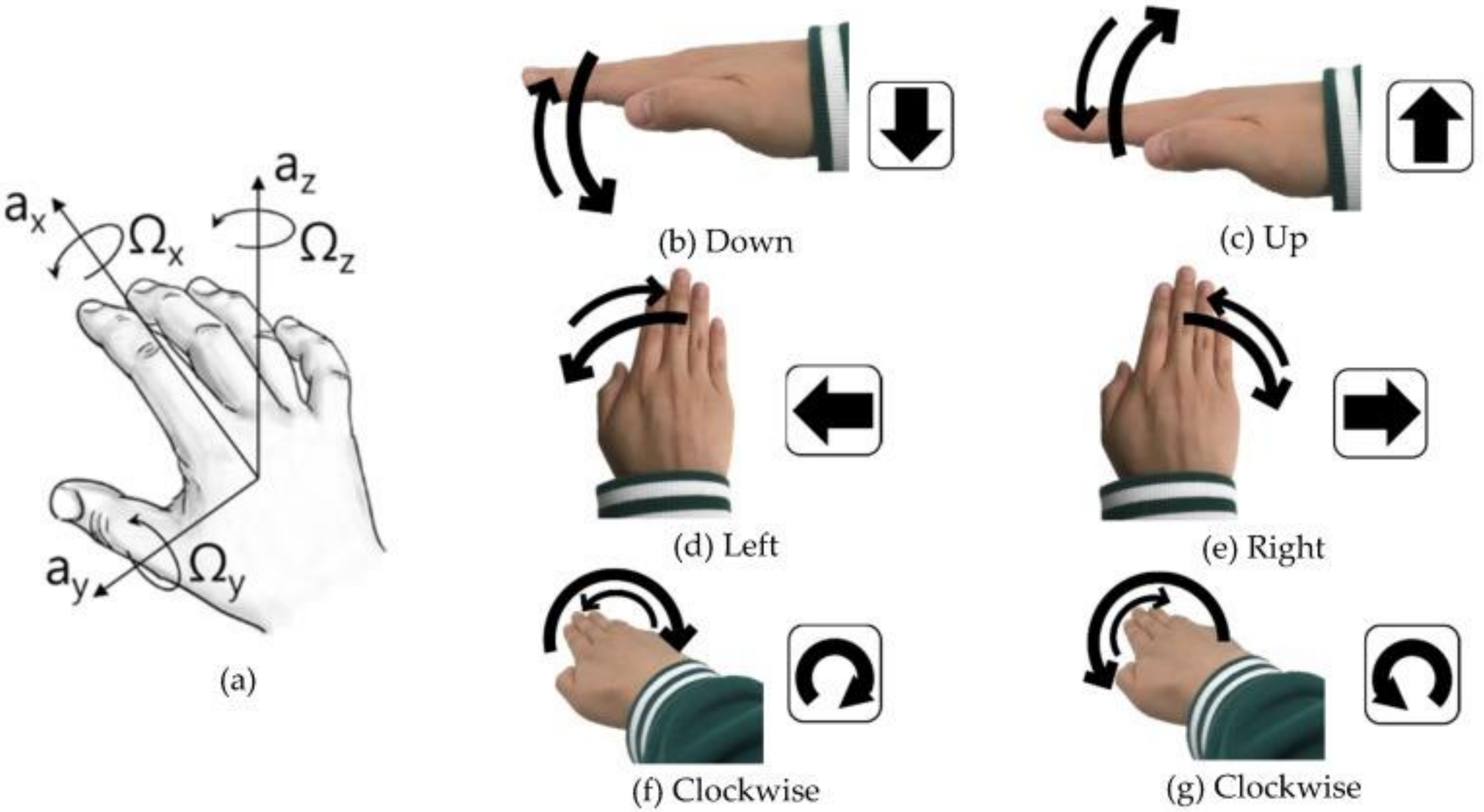
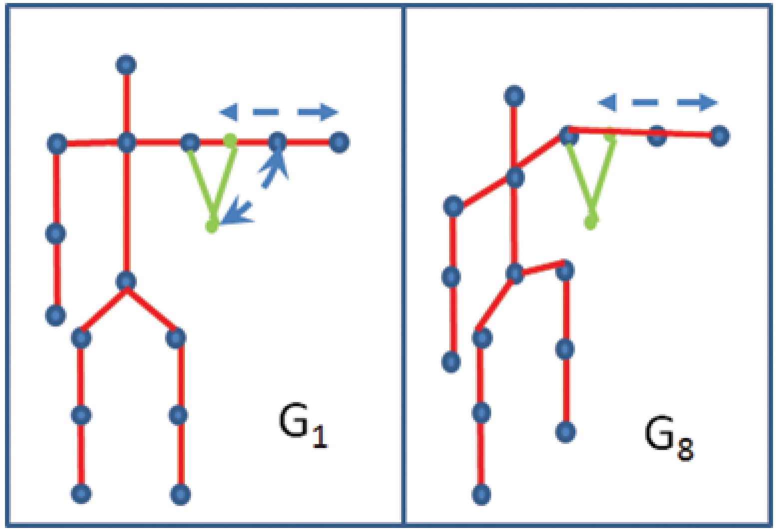
Gesture Recognition (1)

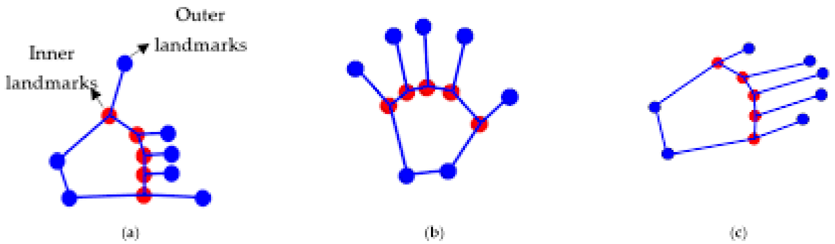


Gesture Recognition (2)



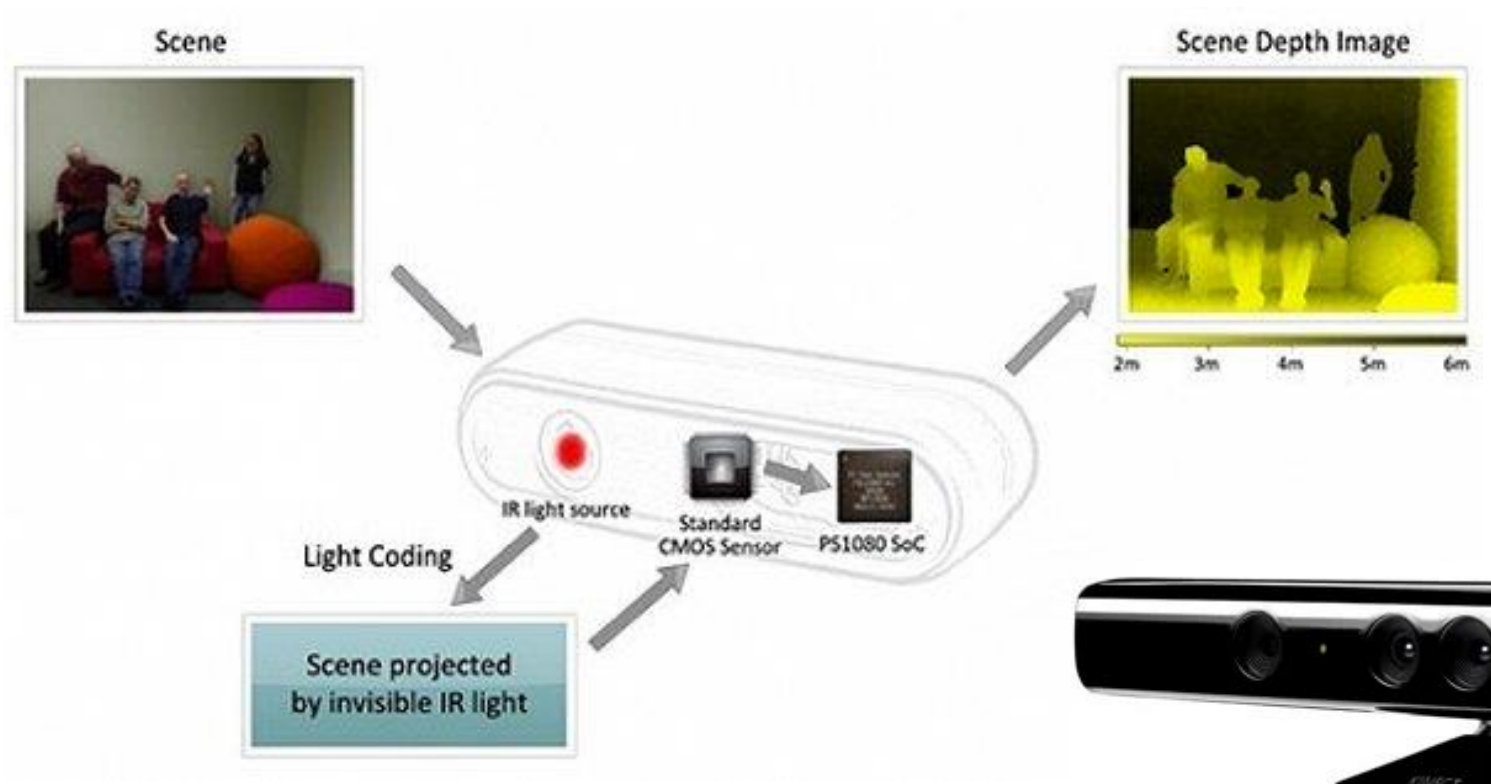
How to recognize?





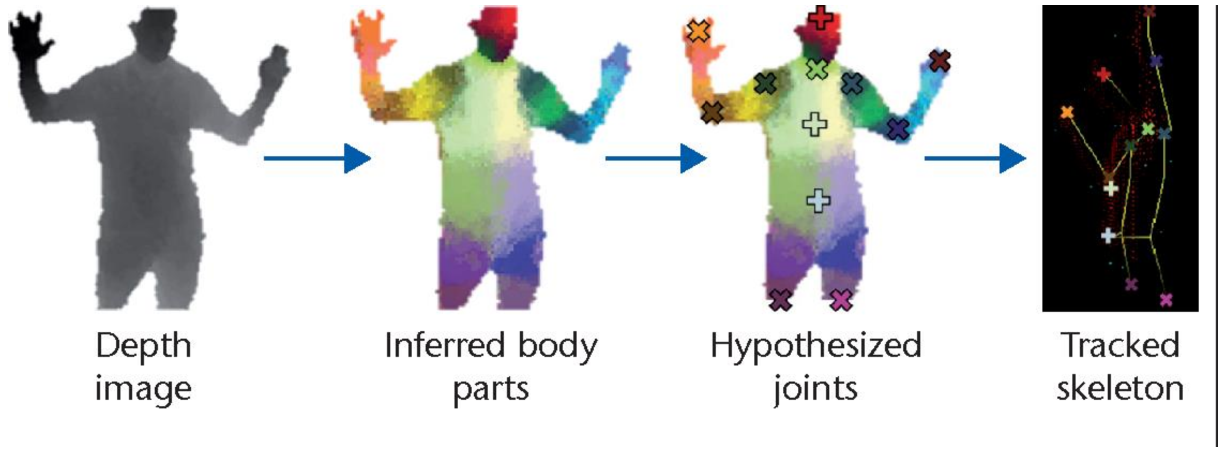
Controlling Drone with Gestures

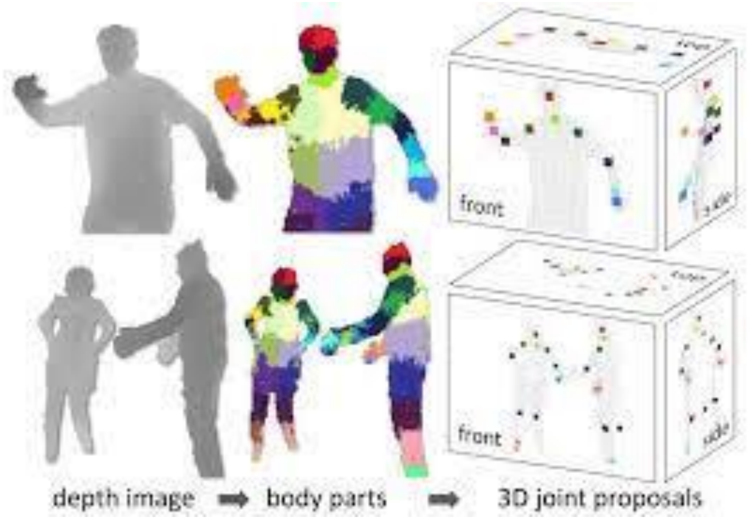
Kinect


Scene
Scene Depth Image
2m
3m
4m
5m
6m
IR light source
Standard
CMOS Sensor
PS1080 SoC
Light Coding
Scene projected
by invisible IR light

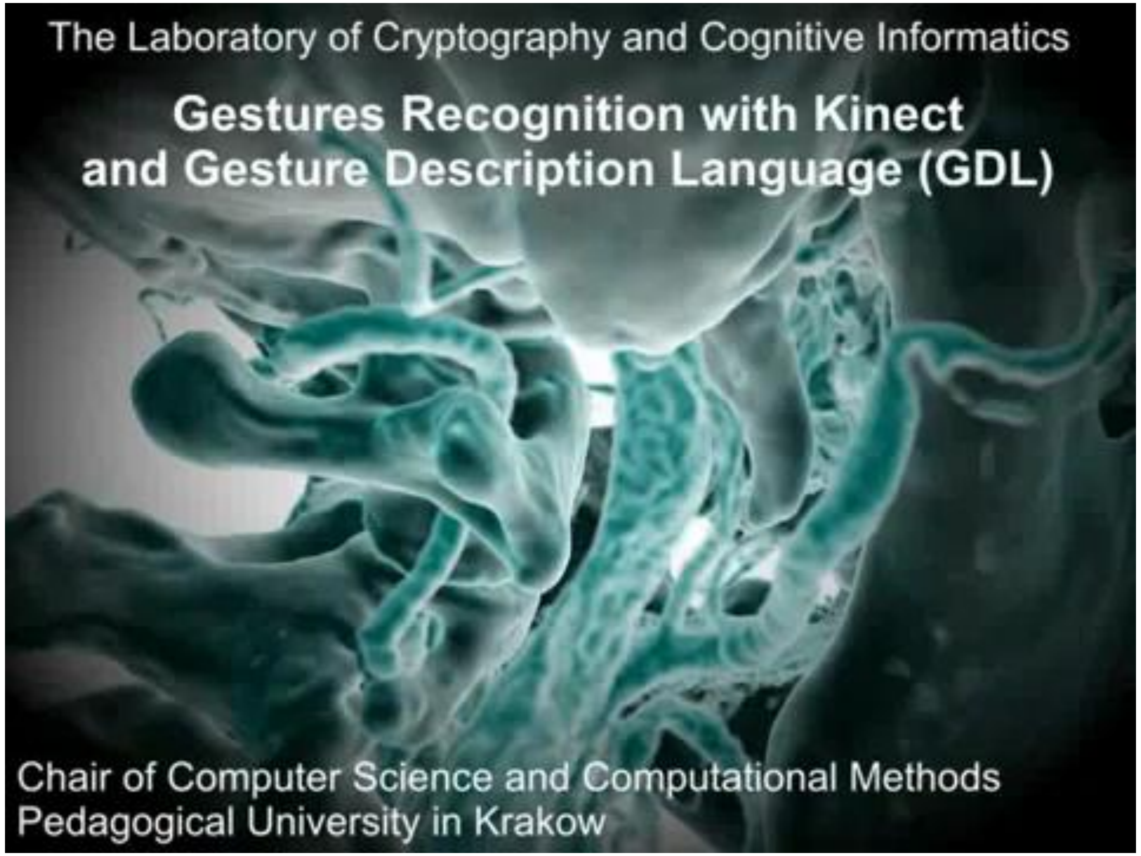

XBOX 360

Depth image?


Depth image
Inferred body parts
Hypothesized joints
Tracked skeleton

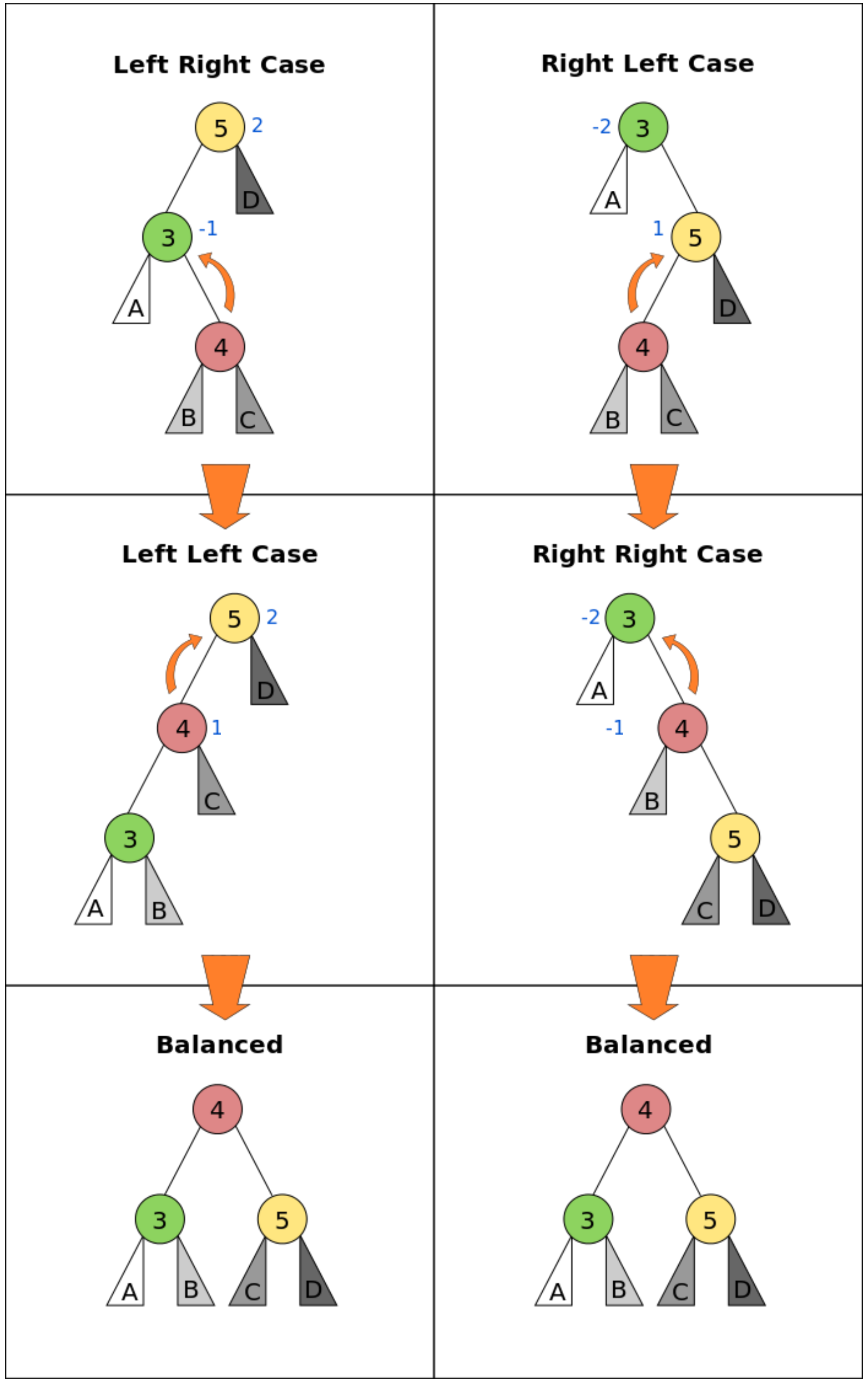

front
side
front
side
depth image
body parts
3D joint proposals

Video


The Laboratory of Cryptography and Cognitive Informatics
Gestures Recognition with Kinect
and Gesture Description Language (GDL)
Chair of Computer Science and Computational Methods
Pedagogical University in Krakow

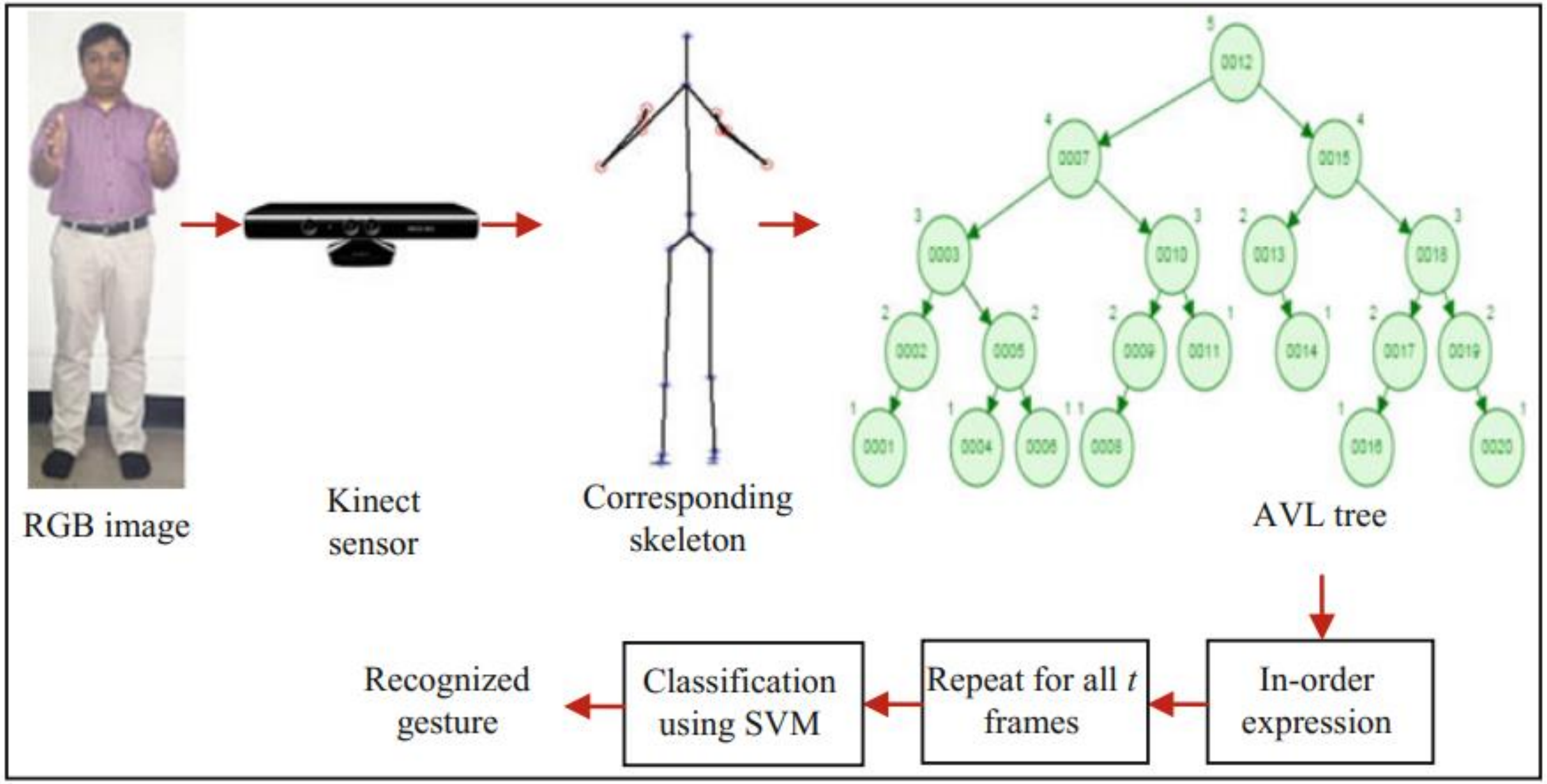
AVL Trees

- AVL tree is a self-balancing Binary Search Tree (BST) where the difference between heights of left and right subtrees cannot be more than one for all nodes. The above tree is AVL because differences between heights of left and right subtrees for every node is less than or equal to 1.



AVL for gesture detection?

- Human body structure is balanced with head as a parent node and all other parts as its children nodes!



How it works?

- They draw a balanced binary tree from 20 joints where they have empirically adjusted weightage value to the 3D coordinates.

- The weightage value assigned to each coordinate is based on the contribution of that coordinate to impart fruitful information so as to recognize the translation of the joints from previous frames.

- Thus, appropriate weights need to be given to the x, y and z coordinate values, and weighted sum of these coordinates is taken as an input to form a node of the AVL tree.

More information…

- Body gesture is the most important way of non-verbal communication for deaf and dumb people.
- Microsoft’s Kinect sensor is used to act as a medium to interpret such communication by tracking the movement of human body using 20 joints.
- A procedural approach has been developed to deal with unknown gesture recognition by generating in-order expression for AVL tree as a feature.
- Here, 12 gestures are taken into consideration, and for the classification purpose, kernel function-based support vector machine is employed with results to gesture recognition into an accuracy of 88.3%.